No. of Functional Features Included in Project

Date	21 October 2026
Team ID	TDS-AI-2026
Project Name	Personalized Travel Itinerary Planner
Maximum Marks	5 Marks

Project Title

Personalized Travel Itinerary Planner

Feature Summary

Total Functional Features: 22

Feature List

Authentication & User Management (3 Features)

#	Feature	Description
1	User Registration	Sign up with email, password, and display name via Supabase Auth
2	User Login	Secure login with email/password, JWT session management
3	Auto Profile Creation	Database trigger automatically creates a profile record on signup

Trip Management (5 Features)

#	Feature	Description
4	Create Trip	Create a trip with destination, dates, budget, currency, group size, style, interests, notes
5	List Trips	Dashboard view showing all user's trips with estimated costs vs. budget
6	View Trip Detail	Full day-by-day itinerary view with activity cards
7	Update Trip	Edit trip metadata (title, destination, dates, budget, etc.)
8	Delete Trip	Remove a trip with cascading delete of all days and activities

AI-Powered Generation (4 Features)

#	Feature	Description
9	AI Itinerary Generation	Generate complete multi-day itinerary using Gemini LLM
10	LLM Fallback	Automatic failover from Gemini to Groq when primary LLM fails
11	Day Regeneration	Regenerate a single day's activities without affecting other days
12	Duplicate Avoidance	Regenerated days avoid repeating activities from other days

Activity Management (4 Features)

#	Feature	Description
13	Edit Activity	Modify activity type, title, description, location, times, and cost
14	Delete Activity	Remove individual activities from a day
15	Drag-and-Drop Reorder	Reorder activities within a day using drag handles
16	Time Slot Swapping	When reordering, time slots are reassigned to maintain schedule logic

Budget & Cost Tracking (3 Features)

#	Feature	Description
17	Per-Activity Cost Estimation	AI assigns estimated costs to each activity
18	Per-Day Cost Computation	Day costs are dynamically computed from activity costs
19	Trip Total Cost Tracking	Overall trip cost displayed relative to budget on dashboard and detail pages

Security & Access Control (3 Features)

#	Feature	Description
20	JWT Authentication	Bearer token verification using Supabase JWKS with key rotation support
21	Row-Level Security	14 RLS policies ensuring users can only access their own data
22	Per-User DB Scoping	Every database query uses a Supabase client scoped to the authenticated user's token

Feature Categorization

Category	Count	Percentage
Authentication & User Management	3	14%
Trip Management (CRUD)	5	23%
AI / Generative AI Features	4	18%
Activity Management	4	18%
Budget & Cost Tracking	3	14%
Security & Access Control	3	14%
Total	22	100%